

Innovative applications of microbial induced carbonate precipitation (MICP) for extraterrestrial construction

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ABSTRACT

The idea of in-situ resource utilization (ISRU) for building extraterrestrial bases on the moon and Mars has become increasingly popular. This review outlines the latest developments in fabrication of space construction materials using microbial induced carbonate precipitation (MICP) technology. This innovative approach uses the urease of *Sporosarcina pasteurii* to promote the precipitation of calcium carbonate from external carbon and nitrogen sources. By using lunar and Martian stardust (afterward called soils in this review) as raw materials, this method presents a sustainable alternative to the difficult and costly process of transporting building materials from Earth to space. The Moon is rich in ilmenite, and TiO_2 is sometimes envisioned to produce H_2 , O_2 , and H_2O , which could serve as a new water source for MICP on the Moon (Technology Readiness Level ≤ 3). This paper focuses on three key aspects of space construction-related materials and technologies. First, it elaborates the continuous advancement in the understanding of lunar and Martian soils, covering the detailed composition and sourcing methods of lunar soil simulants and Martian soil simulants. Second, it explores the complex operational mechanism and considerable application potential of microbially induced calcite precipitation (MICP) technology. Third, it discusses the improvement to the physical properties of space bio-bricks, which endows them with promising application prospects as space construction materials. Additionally, the paper identifies current challenges in this field, such as technical bottlenecks, and outlines the essential research progress required in the future. It expresses optimism that MICP technology will play a pivotal role in revolutionizing space construction by developing robust and eco-friendly materials.

1. Introduction

The Moon is the closest celestial body to Earth. Human exploration of the Moon has evolved alongside advancements in science and technology, from Soviet cosmonaut Yuri Gagarin's historic orbital flight around Earth in 1961 to American astronaut Neil Armstrong's iconic first steps on the lunar surface in 1969 [1,2]. This progression has also been

marked by significant achievements from China, particularly through the Chang'e lunar probes and the Yutu lunar rover [3]. In 2020, China's Chang'e 5 probe successfully landed on the designated site on the lunar nearside, representing the country's first mission to collect samples from an extraterrestrial body. That same year, the National Aeronautics and Space Administration (NASA) entered into the Artemis Accords with space agencies from Australia, Canada, Italy, Japan, Luxembourg, the

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United Arab Emirates, and the United Kingdom, signaling international collaboration for lunar exploration, with a projected crewed return mission initially scheduled for 2024 [4]. Furthermore, China has outlined plans for a lunar research station, with ambitions to establish a permanent lunar base by the early 2030s (see Table 1).

A major barrier to lunar base construction is the exorbitant cost of transporting materials from Earth to the Moon. Duke et al. (2003) estimated this cost at ~\$90,000 per kilogram [5], while a 2016 NASA survey found sending one pound (~454 g) of material to lunar orbit costs ~\$10,000 [6]. Though technological advances are reducing transportation costs, the savings remain insufficient for large-scale base development—transport still constitutes a significant portion of lunar infrastructure expenses. To address this, reducing reliance on Earth-based transportation and leveraging lunar resources have become imperative [7], with in-situ resource utilization (ISRU) emerging as a key solution. ISRU encompasses the extraction, transformation, storage, and use of extraterrestrial materials (e.g., lunar regolith, water ice) during space missions, and the Moon's surface—covered in regolith and meteoric debris (rock fragments, powders, molten glass, etc. [4,7])—rich in SiO₂, Al₂O₃, and CaO (similar to Earth concrete)—provides abundant potential building resources for ISRU. Lunar soil samples (13 kg from Apollo 11 pre-1969; ~115 kg from all Apollo missions by 1972; 321 grams collected between 1970–1976) and simulants have enabled researchers to characterize soil properties (composition, particle size, shear strength, compressive strength). Further, water resources critical for ISRU have been confirmed: Feldman et al. [5,8] analyzed Lunar Prospector neutron data to hypothesize water ice in polar permanently shaded regions (PSRs, high hydrogen concentration, low diffusivity); India's Chandrayaan I mission directly detected water vapor in the lunar atmosphere and confirmed high-latitude subsurface ice via its Moon Impact Probe (MIP) and Altitudinal Composition Explorer (CHACE) [9]; Chang'e 5 samples later provided data on water distribution across lunar latitudes [10,11].

Three main types of lunar construction materials have been developed: sintered materials, 3D-printed structures, and bio-bricks. Sintered materials (e.g., calcium aluminate cement) require high temperatures (>1000 °C). For instance, producing calcium aluminate cement involves heating mixtures to 1527 °C (to separate non-cement components) and then 1727 °C for final curing [12]; Han et al. [13] found optimal sintering temperatures for HUST-1 lunar simulant to be 1050–1100 °C (atmospheric pressure) and 1000–1050 °C (vacuum), achieving 58.45–67.68 MPa compressive strength after 20 kN pre-compaction and 2-h sintering at 1050 °C (25 kPa vertical pressure). Hybrid microwave sintering (volumetric heating via electromagnetic energy [14]) and

solar-focused heating align with ISRU but face challenges (unstable sunlight, need for further heat concentration, unresolved technical barriers [15]). 3D printing (Additive Manufacturing) can create bacteria-loaded microgels that form CaCO₃-containing biomineral composites with 3.5 MPa compressive strength [24,25], but like sintering, it lacks the environmental and cost advantages of bio-bricks.

In contrast, bio-bricks—reinforced lunar regolith bricks produced via Microbially Induced Calcium Carbonate Precipitation (MICP)—avoid high-temperature calcination. MICP uses urease-active bacteria (e.g., *Sporosarcina pasteurii*) to catalyze CaCO₃ precipitation when a cementing solution is introduced [16]; bacteria (e.g., *Bacillus*) produce urease to hydrolyze urea into CO₂ and NH₄⁺ [17]; CO₂ dissolves in water to form CO₃²⁻, which reacts with dissolved calcium salts to precipitate low-solubility CaCO₃, binding lunar soil particles and solidifying the substrate [16]. As the economy expands, the application of biotechnology continues to advance, particularly in the areas of lignocellulose biodegradation and its conversion processes [136,139]. Biochemical advantages include low cost and environmental friendliness (bacterial mineralization reduces soil permeability [16]). Related optimizations include: using limestone waste and acetic acid as a low-cost calcium source [18]; adding peptides to enhance MICP-consolidated sand strength [19]; optimizing bacterial ratios (0.25 wt% *Bacillus pasteurii*, 0.5 wt% *Bacillus sphaericus*) in cement mortar [20]. Ammonium ion byproducts (a potential pollutant) can be removed via bio-electrochemistry [21] or MICP-zeolite technology [22]; human urine (containing urea and water) can serve as a dual MICP resource, with calcium hydroxide added to delay urea hydrolysis [23]. Dikshit et al. [26] demonstrated MICP production of bricks from lunar/Martian soil simulants, achieving a maximum compressive strength of 5.65 MPa with 1 % guar gum and 10 mM NiCl₂.

This paper explores the feasibility of applying Microbially Induced Calcium Carbonate Precipitation (MICP) to lunar and Martian base construction, with specific focus on three core aspects: first, examining the compatibility of in-situ characteristics (e.g., composition, water ice resources) of lunar and Martian soil with the MICP process; second, investigating the survival status and urease activity of functional microorganisms under extreme extraterrestrial environmental conditions (including vacuum, temperature fluctuations, and radiation); and third, exploring the optimization of physical properties of MICP-derived bio-bricks and the integration pathway of in-situ resource utilization (ISRU) resources. Ultimately, this work aims to provide theoretical and technical references for the development of green-building-material schemes tailored to lunar and Martian bases.

2. MICP feasibility in lunar and Martian surface environments

The Moon is Earth's closest and only natural satellite, with a travel time of approximately three days to reach it, compared to nine months for a journey to Mars. Scientists consider the Moon to be an ideal starting point for the exploration of outer space. It could also serve as a staging platform for further space exploration, providing resources for subsequent missions. Mars is considered a potential candidate for human habitation and may support human life and activities [27]. The suitability of astronauts' protective equipment for the surface environments of the Moon and Mars is a critical factor in planning crewed landings. Similarly, the development of building materials adapted to these environments will be essential for ensuring the feasibility of human habitation and production on both celestial bodies.

2.1. The feasibility of MICP on lunar surface

The particles of the weathering layer on the surface of the Moon result from the sustained impact of meteoroids and meteorites over billions of years. This continuous bombardment, along with cosmic and solar radiation, has contributed to the homogenization of the weathering layer.

Table 1

National timetables and plans for base construction on the Moon and Mars.

Country	Space Agency	Lunar base		Mars base	
		Time	Plan	Time	Plan
China	China National Space Administration (CNSA)	2028s	Lunar research station	2033s	Human Mars Exploration Program
America	National Aeronautics and Space Administration (NASA)	2024s	Artemis Program	2031s	Human Mars Exploration Program
Russia	Roscosmos	2030s	Permanent manned base	2050s	Mars research station
Europe	European Space Agency (ESA)	2030s	Permanent manned program		
India	Indian Space Research Organisation (ISRO)	2029s	Manned lunar landing program		

The temperature at the Apollo landing site varied between -171°C and 111°C [28]. Due to the absence of an atmosphere, the temperature differential between day and night can reach as much as 300°C (see Table 2). This absence of an atmosphere leads to significant fluctuations in surface temperature, with pronounced radiation absorption during the day. The temperature variation with latitude and longitude on the Moon exhibits considerable diversity. Therefore, when selecting building materials, it is essential to account for radiation levels. Additionally, the thermal expansion properties of these materials must be compatible with the Moon's unique environmental conditions. During the lunar day, the maximum solar radiation reaching the Moon can be as high as 1358 W/m^2 . Table 2 compares the physical parameters of the Moon and Mars with Earth's. Horvath et al. [29], using data from the Diviner Lunar Radiometer Experiment, determined that the temperature beyond the entrance of the permanently shaded region in Mare Tranquillitatis remains constant at 17°C . Leonardo et al. [142] analyzed radar images of the Mare Tranquillitatis Pit (MTP), which were acquired by the Mini-RF instrument aboard the Lunar Reconnaissance Orbiter in 2010. The research team identified underground caves with tunnels dozens of meters long within the MTP; these caves can serve as shelters to shield against the harsh lunar environment. Since this finding is based on actual radar images captured in-situ, it indirectly confirms that this region can provide protection against the external environment, thereby supporting the cultivation of bacteria. *Bacillus* can grow and reproduce within a temperature range of $10\text{--}50^{\circ}\text{C}$, and the temperature conditions in Mare Tranquillitatis are sufficient to support the growth of *Bacillus*. Bickel et al. [30] developed and validated a novel method for post-processing images of the lunar permanently shaded regions (PSR), allowing for more accurate topography and water ice distribution predictions. This method can significantly aid in identifying suitable locations for future lunar bases.

Lunar base sites are typically proposed for the Moon's south and north poles, where polar day phenomena occur, resulting in a relatively small temperature difference between day and night. According to the latest report [31], the Chandrayaan Surface Thermophysics Experiment (ChaSTE) of the Indian Space Research Organisation (ISRO) detected a temperature of -10°C at a depth of 80 cm beneath the surface of the Moon's South Pole, in contrast to 50°C on the surface. The temperature variation between the Moon's north and south poles ranges from a temperature variation from -10°C to 50°C . Given the presence of craters on the lunar surface, it is considered feasible to situate a lunar base in the shadows of the ring mountains at the poles.

Feldman et al. [32] analyzed hyperthermal neutron data from the Lunar Prospector mission. They demonstrated the presence of ice in the permanently shadowed regions near the Moon's north and south poles, with an average concentration of 1.5 %.

The in-situ detection of gas-phase water by the CHACE instrument aboard the Moon Impact Probe (MIP) of the Chandrayaan-1 mission in 2010 was further supported by infrared spectra obtained from the Moon Mineral Mapper (M3) aboard Chandrayaan-1, which confirmed the presence of water ice beneath the Moon's surface at high latitudes [9].

Table 2
Comparison of physical parameters of the Moon, Earth and Mars.

Parameters	Lunar	Earth	Mars
Surface area[km^2]	37.9×10^6	510.1×10^6	142.8×10^6
Radius[km]	1738	6371	3395
Gravity at Equator[m/s^2]	1.62	9.81	5.02
Escape velocity at Equator[km/s]	2.38	11.2	
Surface temperature range	-173 to 127 -279 to 261 °F	-89 to 58 -128 to 136 °F	-133 to 27 -207 to 80 °F
Seismic energy[J/year]	10^9 to 10^{13}	10^{17} to 10^{18}	
Magnetic vector field[A/m]	0	24 to 56	
Surface atm pressure[kPa]	0	101.3	
Day length[Earth days]	29.5	1	1.027
Sidereal rotation time	27.332d	23.9345h	24.6283h

As shown in Fig. 1, both CHACE and M3 detect an increase in the relative concentration of water ice with latitude, with the highest concentration observed between 60°S and 70°S . Such high-concentration water ice regions can serve as preferred sites for lunar MICP bases. By simply converting lunar water ice into liquid water using in-situ technologies, this can satisfy the requirement for a reaction medium in the MICP process, thereby avoiding the high cost associated with transporting water resources from Earth.

The exploitation of lunar water ice necessitates the disturbance of the lunar soil; therefore, in-situ sampling and research on lunar water ice must take into account the properties of the frozen lunar regolith during the drilling process. Several countries have initiated efforts to explore underground water ice in the lunar polar regions for sampling and research, including China's Chang'e 7, the U.S. VIPER mission, and Russia's Luna-27 [33]. The mechanical properties of icy lunar regolith (ILR) under extreme environmental conditions play a crucial role in the success of water ice extraction.

The core reactions of MICP technology—urease-catalyzed urea hydrolysis and calcium carbonate precipitation—require liquid water as an essential medium. On one hand, water is a prerequisite for bacterial metabolism, including urease synthesis and the maintenance of cellular activity. On the other hand, calcium sources (e.g., CaO in lunar soil) need to react with CO_3^{2-} generated from urea hydrolysis in an aqueous environment to produce CaCO_3 ; this CaCO_3 then acts to cement lunar soil particles, forming bio-bricks. Therefore, the distribution characteristics of lunar water ice directly determine the "in-situ water accessibility" for MICP technology.

Julie et al. [140] employed three-dimensional thermal and gas diffusion models to investigate the process of extracting water ice from lunar regolith via heating. The research team concluded that the volume constraints of the heating wall require optimization to enhance extraction efficiency. Zhang et al. [141] summarized the following urgent issues to be addressed in thermal extraction.

- (1) Heating efficiency: Direct heating or heat-conducting rods form thermal insulation layers due to lunar soil's low conductivity (rods also affected by dry layers, requiring removal mechanisms); microwave heating avoids insulation via soil penetration but complicates mine structure modification.

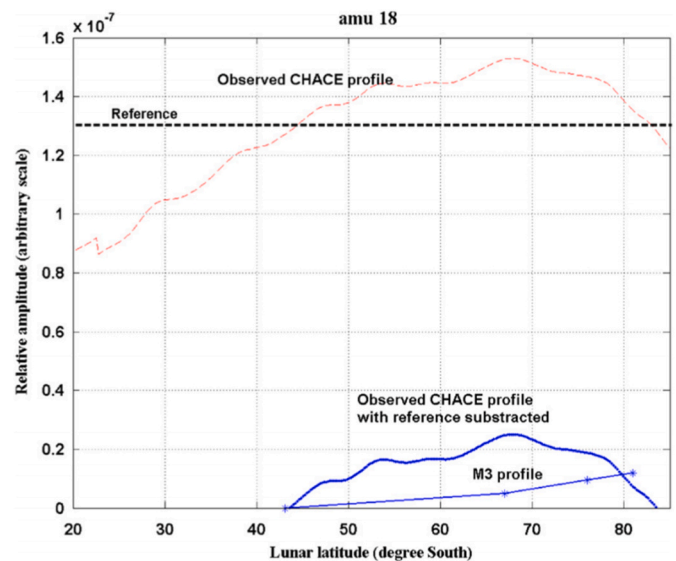


Fig. 1. Schematic of the variation in water ice concentration detected by the Chandra Vertical Composition Explorer (CHACE) and Moon Ship Mineral Mapper (M3) at different lunar latitudes [9].

- (2) Remote energy transmission: Concentrated sunlight is simple for direct heating; lasers/microwaves offer flexible transmission, though microwave beam targeting quality needs improvement.
- (3) Long-distance efficiency: Current wireless power transmission is only short-distance tested; long-distance (lunar crater-scale) efficiency requires further verification.

As shown in Fig. 2, icy lunar regolith (ILR) differs from terrestrial permafrost in its microcomposition, exhibiting irregular particle geometry and the absence of unfrozen water. At -163.15°C , ILR consists of topsoil particles, ice, and pores, whereas permafrost contains more unfrozen water. The thawing of water in permafrost alters the friction, cohesion, pore water pressure, and capillary tension of the particles, leading to distinct mechanical behaviors between ILR and terrestrial permafrost. These differences significantly influence the forces required to break the ice lunar regolith layer during lunar water-ice extraction [33]. Liu et al. [34] analyzed the effects of mineral ratio, temperature, dry density, and moisture content on the unconfined compressive strength (UCS) of icy lunar regolith (ILR). They developed a regression model incorporating these four factors, which revealed that the impact of raw material ratios on the UCS of ILR was significantly smaller than that of temperature, moisture content, and dry density. Additionally, the UCS was found to be relatively unaffected by temperature variations below 180°C . The UCS exhibited a nonlinear increase with the rising dry density of the sample. Jin et al. [35] identified a hydrated mineral $[(\text{NH}_4)\text{MgCl}_3 \cdot 6\text{H}_2\text{O}]$, in the Chang'e-5 lunar soil samples. Water constitutes 41 % by mass, and the presence of ammonium salts could serve as a potential source for urea collection. Bao et al. [36] proposed an inflatable arch structure that creates a temporary indoor environment capable of reducing temperature fluctuations, with the internal pressure reaching 1 atm. In this stable environment, liquid water does not evaporate rapidly. This inflatable arch structure offers a promising direction by providing a stable water source that can support the cultivation of bacterial strains.

Lunar soil can be classified into two groups based on their composition: upland soils formed from plagioclase bedrock and those formed from basalt bedrock [37]. Fig. 3 shows the SEM morphology of lunar soil simulated by basalt and plagioclase. Basalt is the main component of the lunar crust. Mineral composition and temperature affect crustal activity in shallow layers of basalt. Cao et al. [38] found that an increased obsidian content in basalts correlates with a higher likelihood of shallow lunar earthquakes in these regions.

Samokhvalov [49] found that codoped N-TiO_2 can be an effective photocatalyst for hydrogen production, as shown in Equation (1).

N-TiO_2



Lunar highland soils (rich in aluminum and calcium) naturally contain ilmenite, which serves as the primary source of titanium dioxide

(TiO_2)—consistent with the 0.6 wt% of TiO_2 content in the LHS-1 lunar simulant (see Table 4). This enables the in-situ synthesis of nitrogen-doped titanium dioxide (N-TiO_2 , corrected from "titanium nitride") without the need for transportation from Earth, aligning with MICP's goal of reducing reliance on Earth-derived materials.

Second, the hydrogen and co-produced oxygen derived from N-TiO_2 meet MICP's operational requirements: H_2 addresses lunar energy instability by powering fuel cells—solar radiation during the lunar daytime drives N-TiO_2 to split polar water ice into H_2 and O_2 , with H_2 stored for nighttime use. This stored H_2 maintains the 28°C optimal temperature required by *Sporosarcina pasteurii* and powers MICP equipment (e.g., peristaltic pumps). Meanwhile, the co-produced O_2 resolves the Moon's oxygen deficit, meeting the requirement for sustaining the aerobic bacteria critical to MICP. However, further discussion is still needed to clarify how this entire process can be stably implemented in lunar environments.

Various space agencies are deploying landers and rovers to explore the Moon. A thorough understanding of the physical and chemical properties of lunar regolith is crucial for the future construction of human exploration bases in outer space. However, the limited capacity of landers and rovers restricts the amount of lunar soil that can be collected, hindering extensive experimentation and research. Lunar regolith is particularly rich in ilmenite, making the production of lunar analogs a key focus for research organizations advancing their scientific programs. For instance, the University of Minnesota developed the MLS-1 analog, a high-titanium basalt similar to that found near the Apollo 11 landing site [39]. JSC-1 is an analog of Apollo 14 basalt, synthesized from volcanic ash with a high glass content [40]. JSC-1A is an analog of lunar weathering processes observed at the Apollo 17 landing site, with a composition resembling the weathered lunar surface [41]. CAS-1 is a lunar soil analog derived from low-titanium basalt clasts in the Changbaishan volcanic region, developed by the Chinese Academy of Sciences for the China Lunar Prospecting (CLP) soft-landing experiment [42]. Table 3 presents the sources and applications of significant lunar highland analogs. Based on the Exolith Lab's Lunar Highlands Simulant (LHS-1) and Mars Global Simulant (MGS-1), both of which have an average particle size of $90\text{ }\mu\text{m}$ (see Fig. 4), it is reasonable to infer that the surface dust of both the Moon and Mars is highly suitable for specific applications. Rong et al. [43], utilizing coupled magnetic nanoparticle-mediated isolation and single-cell image recognition, identified *Bacillus* species with a size range of $5\text{--}10\text{ }\mu\text{m}$, capable of penetrating the dust on both the Moon and Mars to facilitate Microbially Induced Calcium Carbonate Precipitation (MICP), thus enabling soil consolidation. Clogging of equipment due to small particle sizes also occurs during the collection and utilization of lunar/martian dust.

Additionally, smaller volumes of urease can be released into the lunar soil by disrupting bacterial cells. Urease is synthesized in the cytoplasm of bacteria. Song et al. [44] investigated the microstructural distribution of CaCO_3 in MICP-consolidated sandstone and found that the sandstone's porosity and structure influenced the distribution of CaCO_3 . As the sandstone's porosity decreased, the elastic modulus and brittleness index rose significantly, with a maximum increase of 229 %. Hammes and Verstraete [45] proposed that extracellular polymers, cell membranes, and inactive spores can all serve as nucleation sites for CaCO_3 . Zhang et al. [46] found that calcium carbonate precipitation primarily occurred on small spherical aggregates that polymerized into calcium carbonate during the early stages. They also demonstrated that urease can be effectively released through cell fragmentation. On the one hand, this process can reduce the volume of gaps in the lunar soil, facilitating mineralization; on the other hand, it does not inhibit the reduction in calcium carbonate crystal nuclei due to bacterial fragmentation. In our subsequent experimental plan [151], we aim to slow down the MICP process by optimizing the concentrations of the bacterial solution and cementing solution, and to increase the number of curing cycles to facilitate the uniform distribution of calcium carbonate. Ultrasonic disruption of bacteria allows bacterial fragments to penetrate

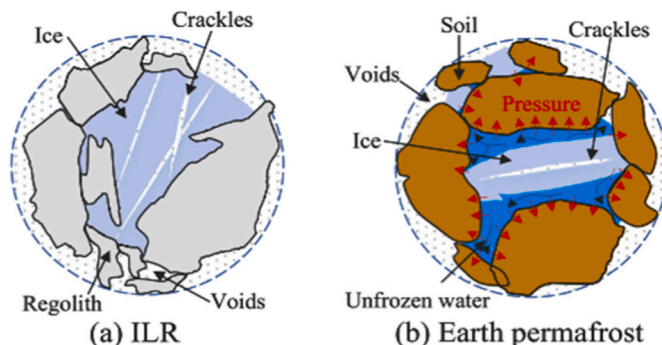


Fig. 2. Icy lunar regolith (ILR) and icy earth regolith [33].

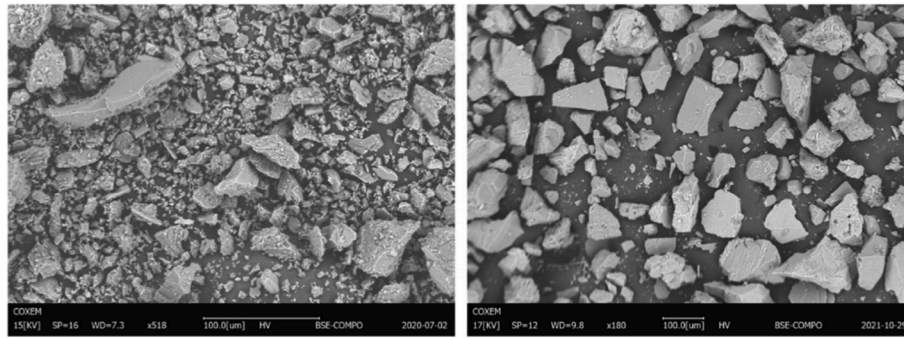


Fig. 3. Electron microscope scans of two types of basic materials of lunar soil simulant. (a) Lunar soil simulated by basalt. (b) Plagioclase-simulated lunar soil [34].

Table 3

Lunar soil simulant representing weathered layers in the lunar highlands [47].

Name	Simulant details	Analogue source	Country	Type	Purpose
MLS-2	Minnesota Lunar Simulant	Duluth Gabbro Complex	USA	Lunar highland	General use
Shawmere	Shawmere anorthosite	Archean Shawmere Complex, Ontario	Canada	Lunar highland	Geotechnical Studies & designing equipment
OB-1	Olivine-Bytownite	Archean Shawmere Complex, Ontario	Canada	Lunar highland	Geotechnical properties & designing equipment
CHENOBI	Shawmere Complex	Shawmere Anorthosite	Canada	Lunar highland	Geotechnical testing
NU-LHT-1 M & 1D NU-LHT-2 M & 2C	NASA-USGS/Lunar Highland Type	Stillwater	USA	Lunar highland	General use
Kohyama base simulant	Kohyama Gabbro base	Kohyama	Japan	Inter mediate of mare & lunar highlands	General use
TUBS-T	TU Braunschweig, Germany	Scandinavian metamorphosed gabbroic complex	Germany	Lunar highland	General use
LHS-1	Lunar Highland Simulant, University Central of Florida		USA	Lunar highland	General use

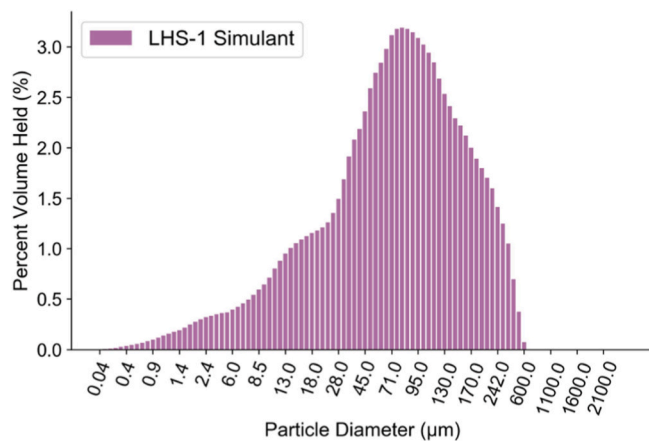


Fig. 4. The distribution chart of the proportion of different particle size ranges of LHS-1(data source: Exolith Lab®)

soil crevices more effectively. Additionally, the smaller volume of EICP (enzyme-induced calcite precipitation) enables it to better mitigate pore clogging.

Smaller pores may become clogged during the MICP process, especially as bacteria secrete urease to induce calcium carbonate precipitation. To address this, strategies such as strain optimization (using smaller-sized bacteria with retained urease activity), regolith pre-processing (removing fine particles to homogenize pore structure), or step-wise injection of bacteria and reaction solutions can be considered. These strategies have been experimentally validated in terrestrial studies and could be adapted for Martian applications.

Oxygen is the most abundant element in lunar rocks, accounting for approximately 44 % of the total weight. It is in metal oxides, including

oxides of magnesium, iron, titanium, aluminum, and silicon. LHS-1 is a lunar highland simulant modeled from Apollo soil data. Table 4 lists the contents of various oxides present in LHS-1. The Moon is rich in ilmenite, primarily in the form of TiO_2 . TiO_2 exhibits excellent photocatalytic properties, facilitating electron transition on its surface under ultraviolet irradiation, which generates electron-hole pairs and thereby promotes REDOX reactions. Hirakawa et al. [48] demonstrated that the surface oxygen vacancies of TiO_2 could facilitate the transformation of water and nitrogen into ammonia through photocatalysis. Samokhvalov [49] found that nitrogen-codoped titanium dioxide can produce hydrogen via photocatalysis. Zhang et al. [50] reported that H_2O can produce CH_4 through the photocatalytic reduction of CO_2 with copper-modified nanopore TiO_2 . TiO_2 holds potential as a resource for in-situ resource utilization (ISRU), offering opportunities for further exploration of production activities on the Moon, which represents one of the key directions for future research ($\text{TRL} \leq 3$).

Table 4

Content of various types of oxides in LHS-1.

Oxide	Wt. %
SiO_2	51.2
Al_2O_3	26.6
CaO	12.8
Na_2O	2.9
FeO	2.7
MgO	1.6
TiO_2	0.6
K_2O	0.5
P_2O_5	0.1
MnO	0.1
LOI*	0.4
Total*	99.4

2.2. The feasibility of MICP on Mars surface

Mars shares a similar environment with Earth and is considered the most Earth-like planet humans observe. As shown in Table 1, the length of a day on Mars is close to that of Earth, with a duration of 24.6283 h, similar to Earth's 23.9345 h. If humans were to land on Mars in the future to establish a base, the planet's natural light cycle would be conducive to human habitation. As shown in Table 1, the surface temperature range on Mars spans from -133°C to 27°C , and specific locations on the planet meet the temperature conditions necessary for sustaining life [51]. According to scientific literature [52], evidence suggests that Mars' surface once contained water. For instance, numerous signs of fluvial erosion are observed on Mars' surface, and a potential water source exists near the equator, which could be accessed by excavating 1–2 m below the surface [53]. Some scholars have hypothesized that the surface of Mars was once warm and humid, as evidenced by the extensive fluvial erosion on the surface, which formed plains, canyons, and valleys [54,55]. A vast ocean may have existed in the northern hemisphere of Mars [56]. Additionally, substantial evidence suggests the presence of water beneath the Martian surface and within caves [57,58].

Water is a non-negotiable prerequisite for MICP. In the context of Mars, while the planet is generally arid, studies have indicated the presence of groundwater, particularly near the equator. Analyses of Mars' surface features and subsurface radar data suggest that there could be regions with relatively accessible water sources. However, before considering these as viable water sources for MICP, a detailed assessment of their characteristics is necessary.

For MICP, the water source should be free of contaminants that could inhibit bacterial activity. Bacteria like *Sporosarcina pasteurii*, commonly used in MICP, require a suitable aquatic environment to maintain their metabolic functions, such as urease synthesis and cell viability. High concentrations of heavy metals or other toxic substances in the groundwater could potentially disrupt these processes. Additionally, the temperature of the groundwater is a crucial factor. The optimal temperature range for the activity of *Sporosarcina pasteurii*, related MICP reactions is typically between 10 and 50°C . If the equatorial groundwater is either too cold or too hot, it may need to be thermally adjusted, which would add complexity to the overall MICP process.

Dubey et al. [152] confirmed that perchlorates inhibit the metabolic and survival capabilities of bioprecipitating microorganisms. To cope with perchlorate stress, these bacteria exhibit "multicellular-like behavior" (forming distinct cell clusters as perchlorate concentration increases) and enhanced extracellular matrix (ECM) release—these phenotypic changes represent compensatory adjustments for metabolic damage (e.g., oxidative stress, enzyme activity inhibition). The Martian surface lacks magnetic field and atmospheric protection, resulting in abundant cosmic radiation (e.g., solar ultraviolet rays, galactic cosmic rays). This radiation can decompose perchlorates into low-toxicity chloride ions (Cl^-) and oxygen (O_2), thereby alleviating the inhibitory effect of perchlorates on microorganisms. This process requires no additional energy input and conforms to the principle of ISRU.

As summarized in Table 5, Martian soils are primarily composed of pyroclastic minerals, constituting 80–85 wt% of the soil due to physical weathering caused by wind, temperature fluctuations, and rock fragmentation due to meteorite impacts [59]. The Martian atmosphere consists primarily of 95 % carbon dioxide, 2.6 % nitrogen, 1.9 % argon, 0.16 % oxygen, and 0.06 % carbon monoxide [60]. Martian soil, mainly composed of basaltic materials, contains various minerals such as iron, magnesium, and titanium-bearing compounds. For MICP, the presence of a suitable calcium source is essential. Although the primary calcium, bearing minerals in Martian soil may not be as abundant as in some terrestrial soils, there are potential sources. Carbon dioxide, the dominant component of the Martian atmosphere, could serve as a carbon source for microbial-induced calcite precipitation (MICP). MICP plays a significant role in bioremediation of toxic compounds and carbon

Table 5

The properties of the soil on the Moon and Mars.

	Lunar regolith	Mars regolith
Formation mechanism	Formed by the continuous impacts of meteorites and meteoroids over billions of years and simultaneously influenced by cosmic and solar radiation, the surface regolith has achieved homogenization.	These features are formed through long-term surface geological processes, such as impacts, wind erosion, water erosion, and glacial modification.
Key minerals and elements	1. Contains ilmenite (mainly existing in the form of TiO_2 , with the TiO_2 content in the LHS-1 lunar simulant being 0.6 wt%); 2. Is rich in metal oxides, with oxygen accounting for approximately 44 % (primarily present in the oxides of magnesium, iron, titanium, aluminum, and silicon); 3. The hydrated mineral $(\text{NH}_4)\text{MgCl}_3 \cdot 6\text{H}_2\text{O}$ (containing water) was identified in the lunar soil samples collected by Chang'e-5.	1. Pyroclastic minerals account for 80–85 wt% of Martian soil, which is formed by physical weathering—including wind action, temperature fluctuations, and rock fragmentation caused by meteorite impacts. 2. It is mainly composed of basaltic materials and contains iron minerals, magnesium minerals, and titanium-bearing compounds. 3. It is rich in iron oxides (which give Mars its reddish appearance), clays, silicates, carbonates, and sulfates.
Particle size range	1–1000 μm	1–1000 μm
Mean grain size	90 μm	90 μm

dioxide sequestration [61,62]. Liu et al. [63,64] employed stable carbon isotope tracking to demonstrate that bacteria in desert soil produced carbonates, with atmospheric carbon accounting for 1.03 % of the calcium carbonate found in the soil sediments. Okyay et al. [62] identified several bacterial genera capable of fixing CO_2 through MICP. Specifically, they discovered that *Sporosarcina*, *Sphingo bacterium*, *Stenotrophomonas*, *Acinetobacter*, and *Elizabethkingia* exhibit high CO_2 absorption efficiency. Scientists hypothesize that hydrogen atoms, influenced by atmospheric waves and sandstorms, undergo thermal escape into space, contributing significantly to the water scarcity on Mars [65].

UV radiation on the Martian surface is a critical factor influencing the survival of microorganisms on Mars. Felipe et al. [154] isolated radiation-resistant bacteria of the genus *Halomonas*. Their research revealed a correlation between radiation resistance and drought resistance: a 5-day increase in drought resistance was associated with an enhanced radiation survival rate at a dose of 1 kGy at room temperature. Godin et al. [143] found that *Bacillus subtilis* exhibited no significant lethality in Martian-like brine formulated with $\text{Fe}_2(\text{SO}_4)_3$, MnSO_4 , and MgSO_4 salts. This suggests that if water exists on Mars, bacteria encapsulated within a liquid protective layer may experience reduced inhibitory effects of Martian radiation, compared to direct exposure to the Martian atmosphere. Whether on the Moon or Mars, the impact of low pressure on microbial cell membranes remains a non-negligible issue—and this could serve as a novel research direction. Research into the effects of ionizing radiation (e.g., gamma rays) on bio-cemented soils has shown that it can impact the microbial viability [160] and potentially alter the treated soil's properties. Furthermore, groundbreaking work performed at the CERN-ISOLDE facility provides direct, quantitative evidence of radiation-induced damage to the crystalline structure of biogenic calcium carbonate itself, offering a mechanistic explanation for the observed mechanical strength loss by analyzing defects in the calcite lattice.

Therefore, the radiation tolerance of these bio-bricks is a key limitation that must be addressed. Future research must prioritize evaluating the radiation tolerance of the optimized mixtures under proton and

gamma-ray irradiation. Mitigation strategies should be explored, including the use of radiation-shielding composites (e.g., incorporating a layer of regolith shielding) and the development of alternative, more radiation-resistant bio-cementation techniques, such as Enzyme-Induced Carbonate Precipitation (EICP).

The reddish appearance of Mars is primarily due to the presence of iron oxide in its soil, which forms rust. Martian soil is also rich in clays, silicates, carbonates, and sulfates. Among the oxides in Martian soil, silica is the most abundant, averaging 45.41 wt% across several physical samples, including those from the Viking 1, Viking 2, Pathfinder, Opportunity, and Curiosity missions. Iron oxides present in the soil include ferric oxide and ferrous oxide. The first three samples listed contain ferric oxide, with an average content of 19.33 wt%, while the latter three contain ferrous oxide, with an average content of 16.73 wt%. Additional oxides in Martian soil include aluminum oxide (9.71 wt%), magnesium oxide (8.35 wt%), calcium oxide (6.37 wt%), and sulfur trioxide (6.16 wt%). The particle size distribution of Martian soils ranges from a few micrometers to several hundred micrometers [59].

In conclusion, while Mars does offer some potential resources for MICP, significant challenges related to water source quality, temperature, and soil characteristics need to be carefully addressed. Future research should focus on in-depth exploration of Martian resources, combined with innovative solutions to meet the specific requirements of MICP technology for successful implementation on the red planet (Mars).

2.3. Progress in the study of building materials for the Moon and Mars

2.3.1. Progress in the study of lunar building materials

According to Goulas et al., the maximum melting temperature of the lunar simulated weathering layer is 1150 °C, while the maximum melting temperature of the Martian simulated weathering layer is 1330 °C [66]. Technologies that involve melting lunar and Martian weathering layer materials for curing must operate at temperatures exceeding these values.

The principle of liquid-phase sintering involves heating raw material powder to a temperature where the low-melting-point particles form a liquid phase. The surface tension of the liquid phase induces the solid particles to rearrange, thereby increasing the density. Unmelted particles and sharp edges dissolve into the liquid phase and reprecipitate, forming a coarser structure with larger, rounded particles. As the densification process slows, the dissolution of particles and their subsequent precipitation and aggregation, along with solid diffusion, contribute to the completion of sintering [67]. Zhou et al. [68] simulated the preparation of polymers using lunar surface simulant soil in a vacuum environment within the temperature range experienced on the Moon. Their method involved synthesizing an alkaline-activated paste by mixing mortar with an alkaline activator (a 8 mol/L NaOH solution), followed by pouring the mixture into a mold and compacting it to await curing. The final curing process resulted in a compressive strength of 38.2 MPa after 72 h.

The weathered layer of the Moon contains sulfur, which can be extracted through heating. Toutanji et al. [69] demonstrated that adding sulfur to concrete increases its compressive strength but reduces its radiation shielding effectiveness. The Moon's surface is characterized by an extremely thin atmosphere, which offers minimal protection from cosmic radiation. Consequently, sulfur-enriched concrete must be thicker than ordinary concrete to provide the same level of shielding under identical radiation intensity. Additionally, Toutanji et al. [69] found that incorporating a small amount of polyethylene can enhance the radiation-shielding properties of the concrete. Bao et al. [36] proposed a lunar architecture featuring an arch-shaped "prefabricated core module, inflatable structure, and in-situ material structure" (Fig. 5). This inflatable arch structure can withstand temperature stresses caused by the nighttime temperature drop, reaching a compressive strength of 9.5 MPa when buried to a depth of 1 m. This capacity allows it to meet the

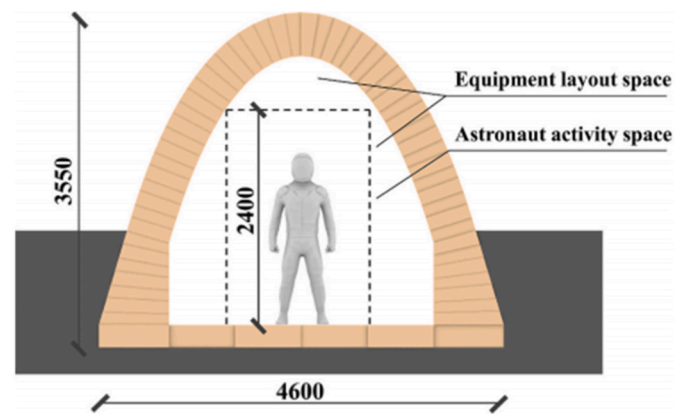


Fig. 5. The standard building unit is 4.6 m wide and 3.55 m high. The upper and sides of the internal space are used to deploy devices. The building unit can meet the needs of three astronauts on short-term missions of seven days [36].

ultimate load-bearing requirements imposed by lunar temperature fluctuations. The structure also provides partial protection against cosmic rays, while its semi-underground configuration, combined with the overlying lunar soil, helps mitigate temperature variations. After exposure to a vacuum environment for 30 days, the modulus of elasticity and tensile strength decreased by approximately 10 %, while the elongation remained unchanged. The compressive strength of the structure can still reach 6.9 MPa.

Zhou et al. [6] developed and characterized a new lunar regolith simulant, Ground Volcanic Scoria (GVS), closely resembling lunar regolith samples from the Apollo missions. After curing for 28 days, the geopolymer achieved a maximum compressive strength of 75.6 MPa and a maximum flexural strength of 6.3 MPa. However, if the curing temperature is too low, the alkali activator will dissolve into the geopolymer gel, compromising its compressive strength. Li et al. [70] prepared polylactic acid (PLA)-lunar surface simulated regolith composites using a 3D printing method, with lunar surface simulated regolith contents of 5 wt% and 10 wt%, respectively. The particle size distribution of the simulated regolith and PLA was consistent. The ultimate tensile strength of the 5 wt% and 10 wt% composites decreased by 14.2 % and 17.6 %, respectively, compared to pure PLA, while the composites exhibited a higher elastic modulus than pure PLA.

2.3.2. Progress in the study of building materials for Mars

The cost of transporting building materials from Earth to Mars ranges from \$10,000 to \$200,000 per kilogram [71], making using in-situ resources from the Martian surface essential. Martian soil is rich in sulfur, with an average sulfur trioxide content of 6.16 wt% based on soil samples collected from Mars [59]. The production of sulfur concrete offers advantages over traditional cement concrete, including low-temperature sustainability, resistance to acidic and saline environments, and recyclability. Additionally, due to the abundance of metals in Martian soils, sulfates and polysulfates may form during high-temperature mixing, potentially contributing to the material's strength [27]. Sulfur was mixed with aggregate at 120 °C during hot mixing, and the monocrystalline sulfur was fully melted and subsequently cooled, transforming into stable orthorhombic polycrystalline sulfur to enhance the stability of the sulfur concrete material. According to Wan et al., the highest compressive strength of 59 MPa was achieved with a sulfur addition of 50 wt% [27].

3. Overview of microbial survival in extreme environments on the Moon and Mars

In the extraterrestrial environment, high vacuum, extreme temperatures, atmospheric anoxia, microgravity, and intense radiation pose

significant challenges to human activities in space and the survival of other organisms. Studies on the survival of microorganisms in space have been conducted, highlighting the difficulties microorganisms face in surviving in such conditions—particularly on the Moon and Mars. This discussion also explores the potential directions for future research, offering insights into the feasibility of Microbially Induced Calcium Carbonate Precipitation (MICP) technology.

The Moon's surface-atmosphere is extremely thin, and the growth and metabolism of *Bacillus pasteurii* require oxygen for support. Copper serves as an excellent active site for CO₂ conversion [72]. Zhong et al. [73] modified lunar soil via copper doping and then applied electrical catalysis to convert CO₂ and produce oxygen on the modified soil. In their study, the oxygen production rate of silicate in lunar soil reached 2.3 mL/min at a current density of 600 mA/cm². Chambers et al. [74] proposed that abundant high-Ti mare basalts and soils on the Moon could release oxygen through hydrogen reduction of ilmenite. Zhao et al. [75] also used laser vacuum pyrolysis TiO₂ to produce oxygen. Additionally, Ross et al. [76] demonstrated the potential of photo-electrochemical (PEC) cells devices to efficiently produce oxygen and fuel under solar radiation conditions on the Moon and Mars.

NASA's Mars Oxygen In Situ Resource Utilization Experiment (MOXIE) successfully produced oxygen from CO₂ in the Martian atmosphere. The experiment compressed the Martian atmosphere into solid oxides, which were then subjected to electrolysis to generate oxygen [77]. Ross et al. [76] also proposed the feasibility of photo-electrochemical (PEC) methods for producing hydrogen and carbon-based fuels. Water ice is present on the Moon and Mars, making water electrolysis a viable option for oxygen production. Additionally, Lomax et al. [78] found that electrolytic oxygen production in lunar gravity was 11 % less efficient than Earth's.

Due to the lack of atmospheric protection on the Moon, solar radiation on the lunar surface is much more intense than on Earth. The most significant concern is the intense ultraviolet (UV) irradiation, which can cause irreversible mutagenic damage to spores through interactions with photons and DNA. Solar UV radiation can be categorized into three broad spectral ranges: UVA (315–400 nm), which accounts for about 6.3 % of the total solar electromagnetic spectrum; UVB (280–315 nm), accounting for about 1.5 %; and UVC (200–280 nm), which makes up about 0.5 %. Although UVB and UVC comprise only about 2 % of the solar spectrum, their intensity and effects are particularly lethal to microorganisms. The lethality of solar radiation in these conditions is exceptionally high [79,80]. DNA is quickly inactivated and induced to mutate due to the strong absorption of UV radiation at these wavelengths. Compounds with saturated bonds are less susceptible to UV radiation damage than conjugated or double-bonded organic compounds with loosely bound and more mobile valence electrons [81]. The average absorption peak of proteins occurs at 280 nm; among amino acids, tyrosine, tryptophan, and phenylalanine are more likely to absorb UV radiation. The average absorption peak of nucleic acids is at 260 nm, where they are particularly prone to mutation and damage. Lederberg [82] proposed a scenario suggesting that life forms whose genetic systems contain more molecules with saturated bonds and fewer with unsaturated bonds would have greater viability in the presence of ultraviolet radiation. In the polar regions of Mars, above 60° latitude, almost no radiation below 300 nm reaches the surface during the winter or summer solstices [83].

Bacillus and its resting spores can survive for several years, primarily due to the DNA of their spores being resistant to alteration. This stability is attributed to the high Ca²⁺-dipicolinic acid content in the spores and the DNA repair mechanisms during spore growth. During spore development, small, acid-soluble sporulation proteins (SASPs) are synthesized and later degraded during spore germination. The α/β -type SASPs are encoded by the monocistronic gene known as *ssp*. [84]. As shown in Table 6, spores of different *Bacillus* species exhibit varying inactivation kinetics under ultraviolet (UV) light, with *Bacillus rodentium* demonstrating the highest resistance [85]. *Sporosarcina pasteurii* is also

Table 6

Resistance of *Bacillus subtilis* cells and spores to moist heat, dry heat, desiccation, and UV rays during growth, modified from literature [84].

Type	Growing cells	Dormant spores	<i>recA</i>	$\alpha^- \beta^-$	$\alpha^- \beta^- recA$
	Wild-type	Wild-type			
Wet heat (90 °C) min	<0.1	18	25	2.5	≤0.5
Wet heat (75 °C) min				100	7
Dry heat (120 °C) min		18	2		
Dry heat (90 °C) min	5			2	≤0.2
Freeze dryings (number)	≤0.2	110	18	7	≤1
UV radiation at 254 nm(kJ/m ²)	36	330	185	18	3

rod-shaped. While specific studies on its UV resistance have not been conducted, it can be speculated that *Sporosarcina pasteurii* may exhibit similar or superior resistance in the harsh conditions of outer space.

Some on-orbit experiments have shown that extreme vacuum drying environments can lead to spore breakage, DNA-protein cross-linking, and other forms of damage [86]. Fig. 7 shows the changes in the survival rate of *Bacillus subtilis* spores over time in dry, hot and humid environments at different temperatures. Miyaji [87] simulated the extraterrestrial environment using machine-generated proton radiation and light from the solar system at 77 K and 4×10^{-6} torr. After exposing *Bacillus subtilis* spores to this environment for 250 years, the survival rate was found to be 45 %. As shown in Fig. 6, spores are significantly more viable in dry heat than in hot and humid environments. This resilience suggests

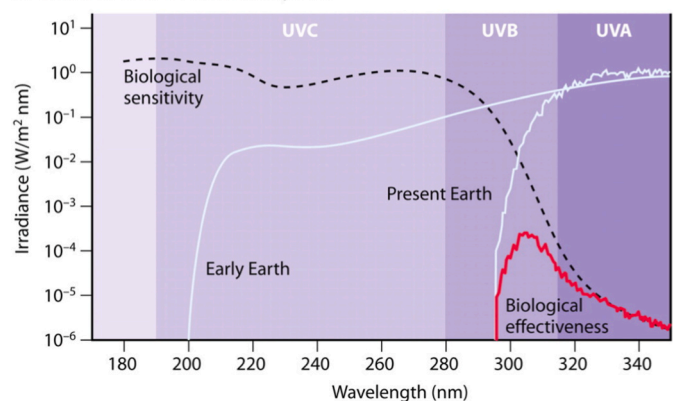
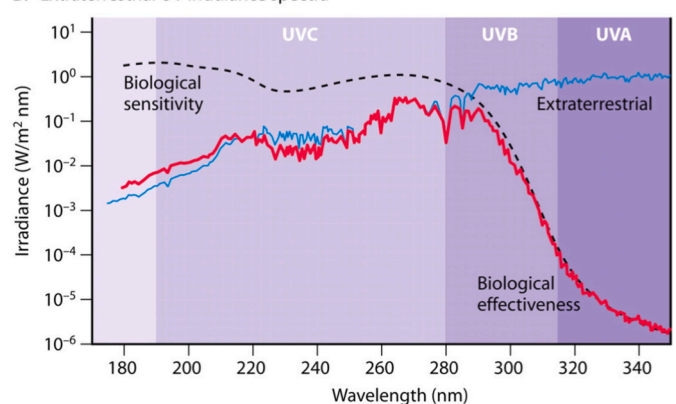
A. Solar terrestrial UV irradiance spectra**B. Extraterrestrial UV irradiance spectra**

Fig. 6. Solar radiation Earth (A) and extraterrestrial (B) UV irradiance spectra, DNA damage action spectra as an example of sensitivity (dashed line), and bioefficacy spectra under Earth and extraterrestrial conditions (bold red line) [79].

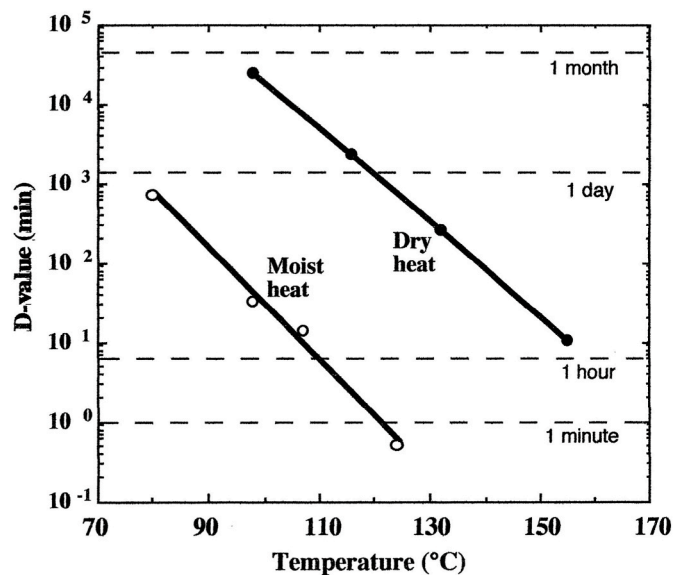


Fig. 7. Variation of spore viability of *Bacillus subtilis* with time at different temperatures in dry, hot, and humid environments [86].

that microbial-induced carbonate precipitation (MICP) could be achievable in space, even without moisture and under the elevated temperatures caused by direct sunlight. It's worth noting that the total incidence of solar radiation in the Martian atmosphere is about 43 % of that found in Earth's atmosphere [88]. Setlow [144] emphasized that spores of the genus *Bacillus* exhibit robust resistance during dormancy. Upon exposure to specific nutrients and other germinants, germination is initiated via membrane-associated proteins, including germinant receptors (GRs), the GerD protein, SpoVA channel proteins, and cortex-lytic enzymes (CLEs). Specifically, GRs perceive nutrient signals, GerD supports GR function, SpoVA channels mediate the release of dipicolinic acid (CaDPA) stored in the spore core, and CLEs degrade the peptidoglycan cortex. Ultimately, these processes enable the spore core to absorb water and swell, thereby completing germination.

Lin et al. [145] investigated the denaturation of proteins by individual spores in a simulated vacuum environment (10^{-5} Pa). The paper identifies α/β -type small acid-soluble proteins (SASPs) as the key factor protecting spore DNA under vacuum: these proteins bind DNA to induce an A-like conformation, shielding it from damage caused by water desorption (a core effect of high vacuum). It also notes that DNA repair pathways (base excision repair, non-homologous end joining) mitigate vacuum-induced DNA breaks.

Massive amounts of CO_2 in the Mars atmosphere make the Martian surface resistant to all radiation below 190 nm. When *Bacillus subtilis* was placed on an aluminum plate and irradiated under these experimental conditions, it took at least 15 min for the number of *Bacillus subtilis* to be reduced to an irrecoverable level. The experiment also showed that the bacterial endospore was not affected by the atmospheric composition in the UV irradiation of the simulated environment on Mars and the monolayer temperature of the bacteria [90]. However, the UV radiation fluxes at different locations on Mars vary. For example, UV radiation fluxes near the Martian equator versus those at the North and South Poles, plateaus versus caves, and places where strains are preserved and cultured can be selected optimally. The inflatable arch structure proposed by Bao et al. [36] is designed to provide partial protection against cosmic rays. Its semi-underground configuration, combined with the overlying lunar soil, effectively mitigates temperature fluctuations. As shown in Fig. 5, this structure can support short-term missions lasting up to 7 days. Laboratory Mars UV simulations [91] demonstrate that *Bacillus* spores exposed to unfiltered Martian UV are inactivated within minutes to hours, with >90 % reduction in

viability within a few hours. Long-term space exposure studies (e.g., EXPOSE-E/ADAPT) show that *Bacillus subtilis* spores exposed to solar UV for 559 days suffered nearly complete inactivation unless shielded, where survival ranged from 8 % to 100 % depending on protective conditions [155]. High-altitude stratospheric balloon experiments further confirm that under Mars-equivalent UV flux, spores are rapidly inactivated within hours [156]. ExoMars RadMap spectrometer data and Curiosity's RAD instrument indicate that equatorial Mars UV flux leads to 90 % spore inactivation within ~ 8 sols, and organic degradation timescales of tens to hundreds of hours, strongly contradicting the notion of multi-year survival without shielding [157].

Xia et al. [92] evaluated the tolerance of phosphate-solubilizing bacteria (*Bacillus mucilaginosus*, *Bacillus megaterium*, and *Pseudomonas fluorescens*) by observing the growth of *Nicotiana benthamiana* in lunar soil simulants. Duri et al. [93] provided a systematic study on the potential sustainability of plants in extraterrestrial environments, focusing on the viability of plant growth in surface weathering layers on the Moon and Mars. Additionally, biomineralization has emerged as a promising concept for resource extraction. Researchers have explored biomineralization to extract rare earth elements under microgravity conditions aboard the International Space Station [94]. Figueira et al. [95] discovered that the filamentous fungus *Penicillium simplicissimum* is capable of extracting metals such as aluminum, iron, magnesium, and calcium from the European Astronaut Center (of ESA)'s Lunar Topsoil Simulant (ESC-1A). Kaksonen et al. [96] discovered that *Acidithiobacillus ferrooxidans* can biologically leach elements such as silicon (Si), manganese (Mn), and magnesium (Mg) from lunar and Martian soil simulants under microgravity conditions. This suggests that Earth-based bacteria could survive in lunar and Martian soils, leveraging their unique biological capabilities to transfer metals and other elements in and out, potentially for human use in extraterrestrial environments. Roberts et al. [97] proposed utilizing human serum albumin (HSA) from astronaut plasma as a bonding agent to bind the weathered layers of lunar and Martian surfaces. This approach could result in compressive strengths of up to 25 MPa. The team also suggested incorporating 3D printing into the process, further enhancing the potential for in-situ building material production in space. A team of NASA Ames Research Center researchers has proposed using commercially produced mycelial materials as bio-composites for potential space applications. Xu et al. [98,99] studied how droplets' contact angle and wetting state change under microgravity compared to Earth's gravity. In the MICP process, bacterial liquids penetrate lunar or Martian soil as droplets, which opens the door for further studies on soil interactions. Melanin-rich fungi have been shown to absorb radioactivity, and using mycelium containing melanin could help shield against a portion of cosmic radiation. Combining such mycelial materials with a weathered layer of extraterrestrial soil could provide a novel source for building materials. Additionally, a study conducted during the Soyuz Mission 8S explored the effects of microgravity on plasmid-mediated gene exchange in bacteria [100]. The comparative test between the Gram-positive bacterium *Bacillus thuringiensis* and the Gram-negative bacterium *Escherichia coli* revealed that the plasmid exchange rate in Gram-positive bacteria increased under spaceflight conditions [111]. This suggests that microorganisms in microgravity undergo genetic mutations, which may impact the performance of MICP. Consequently, when considering MICP for space construction, it is essential to explore how *Sporosarcina pasteurii*, the key bacterium for this process, adapts to or may be genetically modified to thrive in space conditions. Santomartino et al. [153] discovered that for *Bacillus subtilis*, among the three gravity experiment schemes conducted on the International Space Station, there were no significant differences in the final cell count and optical density. This result indicates that by the end of the experiment, the influence of gravity on bacterial growth has been overcome, meaning that microbial-mediated biological production and life support systems can be effectively applied in space (such as on Mars), just as they are on Earth.

Seager [101] proposes that microbial life in space may survive by

encapsulating in droplets, which dry out as the surrounding atmosphere evaporates. The tiny, dried spores enter a dormant phase, during which they can be rehydrated under suitable environmental conditions to resume their activity. Research has also shown that many lichenized fungal and algal cells can withstand harsh conditions in space, including high levels of ultraviolet and cosmic radiation. Specifically, studies have demonstrated that lichens exposed to pre-flight and post-flight conditions exhibit virtually the same photosynthetic activity at different wavelengths of extraterrestrial solar ultraviolet light [102]. Moreover, lichens can recover their metabolic activity within 24 h after vacuum dehydration. Lichens are highly resilient microorganisms that adapt to extreme temperatures, radiation, and dryness. They can survive prolonged periods of desiccation and thrive in some of Earth's most extreme environments, such as high mountains and polar regions. This resilience is mainly due to their unique composition, which includes cyanobacteria (the photobionts) and fungi (the mycobionts). The differentiated cortex of lichens, protective shading pigments, and living or dead cells in the surrounding fungi offer an additional layer of protection. Furthermore, the photoproducts of lichens are well-preserved in space environments because of these adaptive features [103].

4. Research progress of MICP

MICP is an emerging biotechnology applied to combat land desertification and consolidate loose sand. The most extensively studied bacteria for MICP in soils are *Bacillus pasteurii*, *Bacillus sphaericus*, and *Bacillus cereus*, as well as bacteria specific to certain regions, such as *Serratia marcescens* in Brazil, which proves particularly effective for local soils as a strain for MICP-reinforced soil [104]. MICP technology utilizes urease-producing bacteria that catalyze the hydrolysis of urea. These bacteria produce calcium carbonate ions that react with calcium ions to form calcium carbonate precipitates under suitable culture conditions. This process not only solidifies the soil but also reduces its permeability [16]. MICP technology is economically and environmentally friendly from a biochemical point of view, using the mineralization of bacteria to solidify the soil and reduce permeability [105]. MICP has a wide range of applications, including sand column cementation, soil curing, micro-crack repair, restoration of ancient buildings, self-healing concrete [106], sand production control [107–112], consolidated soft coal reservoirs [113].

The mechanism of MICP (Microbially Induced Calcite Precipitation) involves the production of urease by *Bacillus pasteurii*, located in the bacterium's cytoplasm and synthesized as a byproduct of its metabolism. Ma et al. [114] observed that the surfaces of *Bacillus pasteurii*, including the cell walls and extracellular polymers, carry more negative charges than non-mineralized bacteria. This enhanced negative charge allows the bacteria to adsorb more calcium ions (Ca^{2+}) from the surrounding environment. Urease then catalyzes the hydrolysis of urea, producing ammonium ions (NH_4^+) and carbon dioxide (CO_2). The ammonium ions raise the pH, creating an alkaline environment, while the CO_2 dissolves in water and forms carbonate ions (CO_3^{2-}). These carbonate ions combine with Ca^{2+} to form calcium carbonate (CaCO_3), which precipitates around the *Bacillus pasteurii* cells. As a result, the bacteria serve as the nucleation sites for forming CaCO_3 crystals, with *Bacillus pasteurii* acting as the crystal nucleus.

The mechanism behind MICP-driven concrete self-repair involves pre-incorporating dormant microorganisms into the building materials. Upon the occurrence of cracks, moisture, and oxygen within the gaps activate these dormant microorganisms, initiating metabolic activity that generates mineralized precipitates, thereby facilitating the repair of the cracks. Feng et al. [115] designed a method for encapsulating bacteria in calcium alginate hydrogels, where carbon nanotubes (CNTs) wrapped around polydopamine formed a spherical hydrogel, and Calcium SulfoAluminate (CSA) cement uniformly applied a protective shell around the gel. This protective shell ensured that the modified hydrogel remained unaffected in solutions containing high alkaline substances

and concentrations of calcium ions. In the context of geo-energy extraction, "sand production" and "sand control" present significant challenges [110], particularly during the extraction of combustible ice (natural gas hydrate) and other geo-energy sources, such as geothermal heat and hydrogen storage. Qin et al. [116] proposed using MICP to manage sand production during gas extraction from marine hydrate reservoirs, and their study demonstrated the feasibility of this approach. Chen et al. [117] explored corn pulp solution as a more cost-effective nitrogen source than soybean peptone, optimizing the fermentation medium. Their findings showed that this adjustment led to a 24.2 % increase in the urease activity of *Bacillus pasteurii* and a 21.3 % increase in the compressive strength of consolidated aeolian sand. Feng et al. [18] suggested utilizing inexpensive limestone waste and acetic acid reactions as a calcium source, which reduced the production cost of MICP by 31.87 %. This approach not only aligned with sustainable development principles but also expanded the sources of calcium for the process. Shu et al. [118,119] summarized the benefits of incorporating fibers into MICP treatments to improve compressive strength. Their research showed that reducing the roughness of jute fibers enhanced their binding capacity, leading to a compressive strength of up to 3.62 MPa. Song et al. [120] proposed using MICP to control microplastics as a novel approach. Microplastics, small particles resulting from the degradation of plastic waste, pose significant environmental hazards as they infiltrate soils and waterways. Consolidating these particles through MICP represents a promising green environmental protection strategy. Wang et al. [19] demonstrated that adding peptides to sandy soil treated with MICP increased its unconfined compressive strength by 35.03 %. This was attributed to the peptides promoting the activation and growth of bacteria, thus enhancing the overall effectiveness of the treatment. Nasser et al. [20] incorporated a mixture of *Bacillus pasteurii* (0.25 wt%) and *Bacillus sphaericus* (0.5 wt%) into cement mortar. The compressive strength of the treated specimens increased by 28–50 % after 28 days of curing, and their flexural strength improved by 19.29–65.94 % after 120 days of treatment, compared to untreated samples.

Urease-producing bacteria generate significant amounts of ammonium ions when reacting with calcium ions. The discharge of large quantities of ammonium ions may result in water pollution and contribute to eutrophication. Sulonen et al. [21] summarized alternative solutions for recovering ammonia through bioelectrochemical systems (BES), as they demonstrated lower energy requirements. In BESs, the bioelectrochemically generated current drives the transport of NH_4^+ from the wastewater to a concentration chamber through a cation exchange membrane before its subsequent recovery.

Su et al. [22] investigated the application of MICP-zeolite technology for reducing ammonia emissions. Their results showed that the compressive strength of zeolite-treated soil was 0.22 MPa, compared to 0.08 MPa in untreated soil. The study further indicated that zeolite exhibited optimal deamination effectiveness under conditions of 40 °C and pH 9. Similarly, Lin et al. [121] focused on natural zeolite, known for its cation exchange, catalytic, and adsorption properties, and modified with NaCl to enhance its ammonia removal efficiency.

Wang et al. [122] isolated a marine *Bacillus sphaericus* XR1 strain capable of producing both urease and extracellular polysaccharides. The combination of polysaccharide production and calcium carbonate crystallization enhanced the strength and flexibility of the cured materials. This strain selection optimized the brittle properties of MICP-cured materials. Lambert et al. [23] utilized urea from human urine to produce bio-bricks. Calcium hydroxide was added to the urine to stabilize the area and prevent loss. The urea in the urine served as a nitrogen source in the experimental setup, as illustrated in Fig. 8. A peristaltic pump introduced the cementing fluid into the bio-bricks mold. After consolidation, bio-bricks with an unconfined compressive strength (UCS) of 2.7 MPa were obtained, as shown in Fig. 9.

Deng et al. [123] optimized two methods for recovering water and nitrogen from human urine: the High-Temperature Acidification

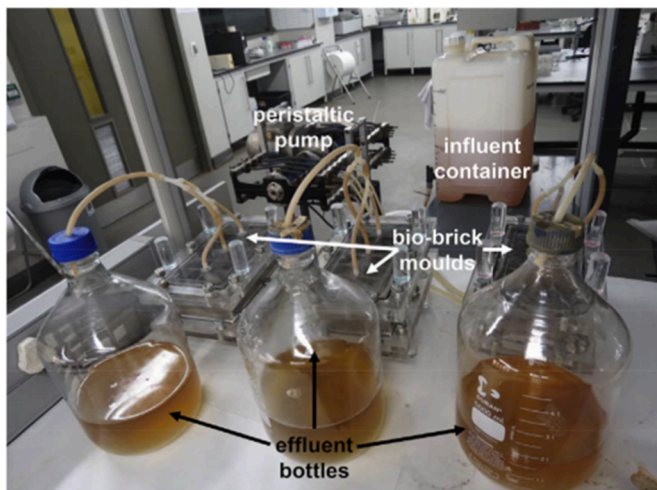


Fig. 8. The setup for the bio-bricks experiment shows the inlet container that holds the cementing solution, the peristaltic pump used to pump the cementing solution into the bio-bricks molds periodically, three bio-bricks molds, and three effluent 5 L glass vials [23].



Fig. 9. Bio-bricks consolidated by urine treatment [23].

Method (HTAM) and the Immobilized Urine Catalyzed Method (IUCM). In both approaches, urine is first hydrolyzed, followed by cooling and collection under alkaline conditions to extract water vapor and ammonia.

In HTAM, urea in the urine is hydrolyzed to ammonia under high temperatures and acidic conditions, and the ammonia and water are then distilled under reduced pressure to maximize recovery. In contrast, IUCM involves the following steps.

1. Dissolving urease powder in a gelatin solution.
2. Condensing and cutting the urease-gelatin mixture into pieces.
3. Crosslinking the mixture with a glutaraldehyde solution.

IUCM exhibits more efficient hydrolysis than HTAM, with moderate reaction temperatures and times. Therefore, it is better suited for recycling human urine in outer space applications.

Enzyme-induced calcium carbonate precipitation (EICP) is a technique that replaces the microbial urea hydrolysis process with enzymes to promote urea hydrolysis. Enzymes in this process can be derived from microorganisms, fungi, and agricultural products, such as soybean urease [127,128]. Compared to traditional MICP, EICP offers several advantages, including lower economic cost, more simple operating conditions, and the absence of the need to cultivate microorganisms. These advantages make EICP a promising approach for specific applications.

Martin et al. [146] modified the EICP technology by incorporating skim milk powder, which significantly enhanced the effectiveness of

EICP in improving the strength of granular materials. This modification was found to promote more calcium carbonate precipitation at the interparticle contact points. Furthermore, they observed that the volume of calcite crystals generated by the modified EICP was larger than that produced by the unmodified EICP. Meng et al. [147] proposed a multi-phase EICP method for soil improvement. For ASTM C778 graded sand treated by this EICP method, its unconfined compressive strength exceeded 10 MPa, with the calcium carbonate (CaCO_3) content being approximately 20 %. Xiao et al. [148] conducted one-dimensional compression tests on quartz sand treated by MICP to evaluate the effects of gradation and CaCO_3 content on the compression behavior of the sand. The study revealed that the compressibility of the specimens increased with an increase in the coefficient of uniformity or a decrease in CaCO_3 content. Additionally, the presence of small particles resulted in more interparticle CaCO_3 bonds; even after the rupture of these bonds, the small particles could still fill the voids between coarse particles. Wang et al. [149] verified the feasibility of applying MICP technology to prevent erosion of granite residual soil slopes. Compared with the untreated soil, the runoff rate of the soil treated by MICP increased by an average of 39.4 %. Huang et al. [150] verified the feasibility of MICP technology for reducing the porosity of granite residual soil under unsaturated conditions. Biological cementation induced by MICP significantly decreased the soil porosity, and the maximum variation in porosity was closely associated with the concentration of the MICP chemical solution and the content of CaCO_3 .

These research findings not only enrich the theoretical framework of biomineralization-modified geotechnical materials but also provide important references for the parameter design and practical application of biocementation technology under different engineering demands (e. g., strength enhancement, compression control, erosion resistance, and pore optimization). Furthermore, they lay a foundation for the subsequent expansion of this field toward complex geotechnical types and extreme engineering environments.

However, one major limitation of EICP is that the enzyme, typically present as free urease in an aqueous solution, lacks sufficient nucleation sites. This deficiency results in a reduced bond strength between the generated calcium carbonate and the medium, which in turn negatively affects the properties of the final cured material. To address this challenge, several studies have explored using exogenous nucleating agents to enhance the process. For instance, skim milk powder, rich in proteins, has been shown to provide effective nucleation sites, thereby improving both the precipitation efficiency and the quality of the calcium carbonate [129]. However, the high cost of skim milk powder restricts its widespread use. In contrast, sucrose and sorbitol, which are inexpensive and readily available, have also been demonstrated to serve as effective nucleating agents for the EICP process [130].

Furthermore, Yan et al. [131] demonstrated the feasibility of using calcium and phosphorus sources extracted from kitchen waste, for example from bones, to induce calcium phosphate precipitation under microgravity conditions. This study highlights the potential of recycling food residues carried by astronauts as a source of calcium and phosphorus, which can be utilized in space environments. This finding opens possibilities for using waste materials to support the development of aerospace bio-based building materials. In conclusion, EICP technology offers a more straightforward and economical alternative to traditional calcium carbonate precipitation methods. Future research should focus on optimizing the provision of nucleation sites to improve the efficiency of calcium carbonate formation and enhance the bonding strength between calcium carbonate and the medium.

5. Progress in bio-bricks research

With the advancement of bioengineering technology and the widespread application of microbial mineralization, the concept of manufacturing bio-bricks using Microbially Induced Calcite Precipitation (MICP) technology has gained significant attention. Bernardi et al.

[124] developed a biologically mediated process for producing bio-bricks, achieving compressive strengths of up to 2 MPa, comparable to bricks produced with cement or lime additives. Bio-bricks are considered more environmentally friendly and sustainable than conventional lime and cement, making them a focus of ongoing research.

Researchers have developed a technique for printing biomaterials in vitro by combining 3D printing with Microbially Induced Calcite Precipitation (MICP) to create porous CaCO_3 -based composites with separation properties [24]. As shown in Fig. 10, lyophilized bacteria are dispersed in a gelatin solution. After cooling to room temperature, the microgel is washed with phosphate-buffered saline (PBS) and then redispersed in a sodium alginate-gelatin solution (Fig. 10a). Calcium and nitrogen sources are incorporated into this bio-containing hydrogel material. The MICP process is then triggered, forming calcium carbonate precipitates within the 3D-printed hydrogel scaffold. This process transforms the soft scaffold into biomaterial composites capable of withstanding compressive forces of up to 3.5 MPa. This method is suitable for remote operation in outer space. However, further research is required to extend this system to produce bio-bricks.

5.1. Progress in the study of lunar bio-bricks

In the context of space exploration and extraterrestrial base construction, conducting biological experiments on bacteria in extraterrestrial environments has become increasingly important. A research team has developed a modular, self-contained lab-on-a-chip (LoC) device designed to cultivate bacteria in microgravity [125]. Fig. 11 presents the technical approach and practical application of the Microbial Analysis in Space (MANAS) modular system based on the self-contained

LoC concept. Fig. 12a shows optical microscope images of rod-shaped bacteria cultured within the prototype device. In contrast, Fig. 12b presents scanning electron microscope (SEM) images of the same bacteria, confirming the optical observations. These results demonstrate the feasibility of cultivating bacteria using the system shown in Fig. 11.

Dikshit et al. [126] demonstrated the feasibility of producing bricks from lunar and Martian soil simulants using the MICP technique. This study used guar gum as an organic polymer, while NiCl_2 was a biocatalyst. Since urease is a nickel-dependent metalloenzyme, increasing the concentration of nickel ions enhances urease activity and accelerates the urea hydrolysis process. Flask growth data indicated that adding 10 mM Ni^{2+} significantly enhanced urease activity, thus speeding up the MICP process. The maximum compressive strength achieved by adding 1 % guar gum and 10 mM NiCl_2 was 5.65 MPa. Chang et al. [23] investigated the effect of xanthan gum biopolymer on soil reinforcement. Xanthan gum fibers directly interacted with the charged surfaces of soil particles, while a matrix resembling hard plastic formed between the uncharged particles. The reinforcement effect was more pronounced in fine-grained soils. High concentrations of xanthan gum in fine-grained soils increased compressive strength and modulus of elasticity due to the strong bonding between xanthan gum monomers and the soil particles. Studies have shown that xanthan gum is most cost-effective when added at a concentration of 1–1.5 %.

5.2. Progress in the study of Martian bio-bricks

Dikshit et al. [26] employed a casting method to prepare bio-bricks from Mars-simulated soil. They achieved a maximum compressive strength of 3.3 MPa by adding 10 mM NiCl_2 and 1 % guar gum. Gleaton

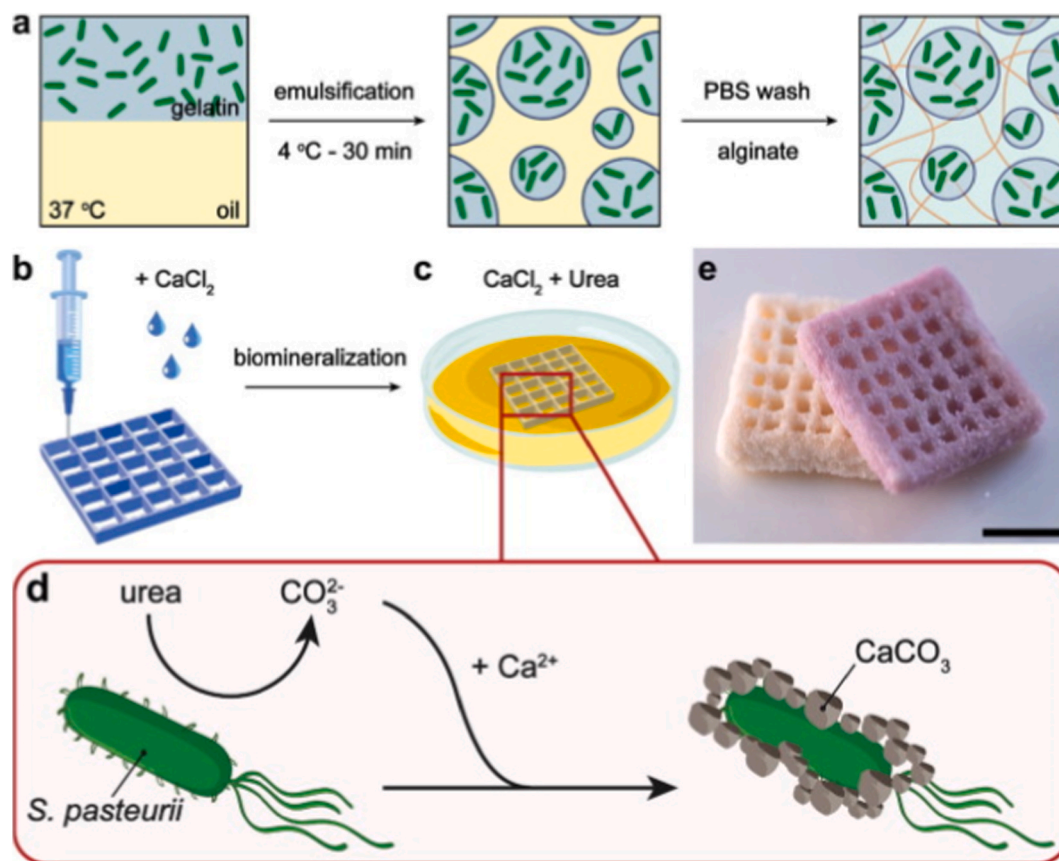


Fig. 10. (a) Diagram of a bacterium binding to a carrier. Emulsify a gelatin solution containing bacteria. Cooling the gelatin solution encased in bacteria becomes a microgel. Wash the microgel to remove oil and active agents from the surface of the microgel. A microgel-containing bacteria is mixed with an alginate solution to form a composite material through 3D printing. (bcd) Add CaCl_2 and urea to trigger the MICP process schematic. (e) Photo of a biomineral composite 3D printed after four days of MICP [24].

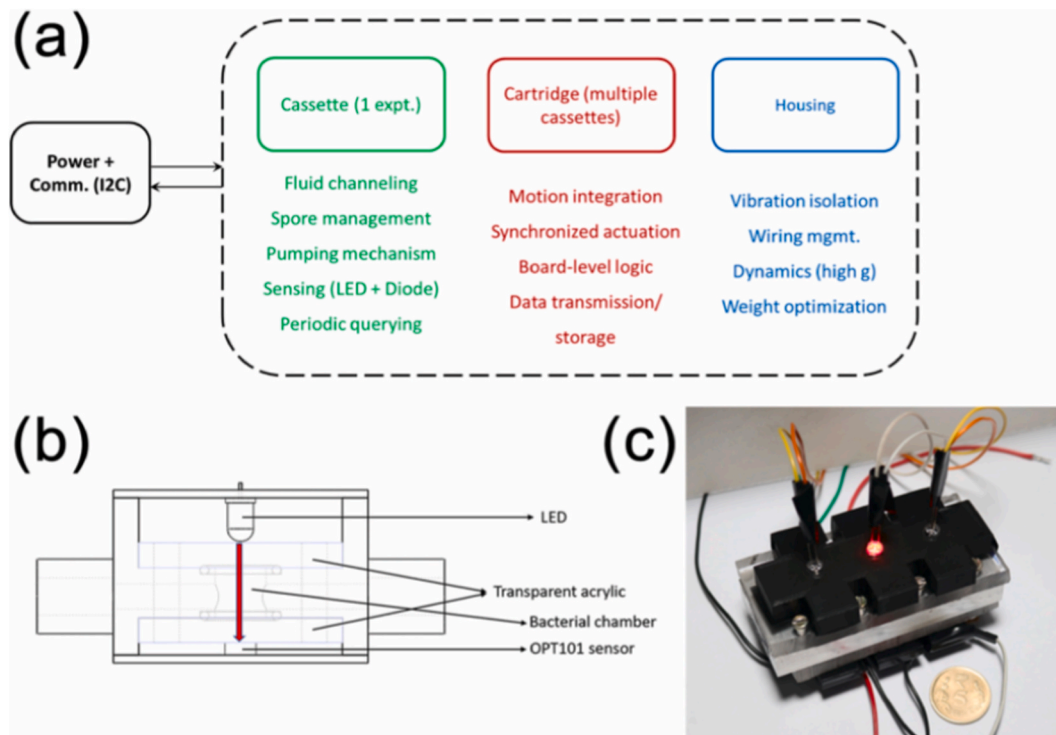


Fig. 11. (a) the modular design of MANAS based on the LoC concept, showing the layered layout of individual cassettes and audio tapes; (b) Schematic of a single box-type internal device containing an LED light source for illuminating the sample of the bacterial chamber with a photodiode (OPT101 sensor) of transmitted light intensity; (c) Outside the individual cassette unit in practice [125].

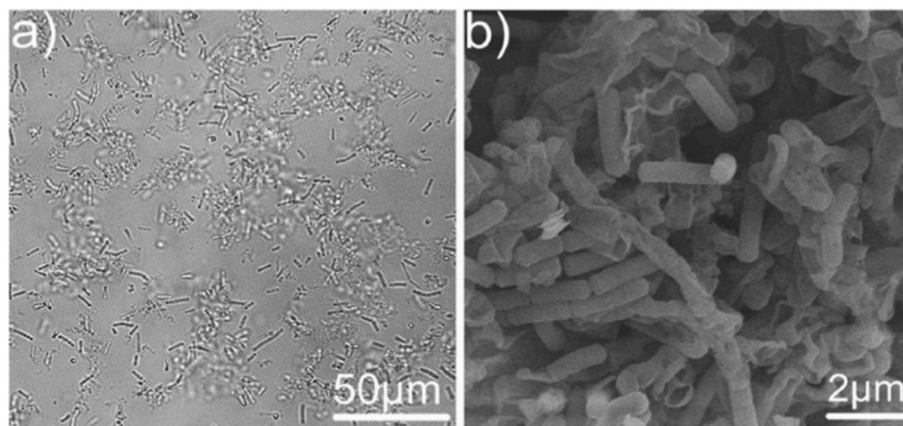


Fig. 12. (a) Light microscopy image after *Sporosarcina pasteurii* cells grow approximately 40 h in the apparatus and (b) scanning electron microscopy image after cell growth [125].

et al. [132] bioconjugated simulated weathered layers on the surface of Mars and investigated the carbonate precipitation produced by the marine microalga *Thraustochytrium striatum*. This microalga, in combination with *Bacillus* sp., was able to grow on urea and acetate. Instead of using calcium chloride, calcium acetate was employed as the calcium source in the bioconjugation process. The bio-grouting method used to produce Martian columns achieved a compressive strength of up to 740 kPa. In a subsequent study [133], the team further examined the effects of particle size and biomass loading on the compressive strength of simulated Martian soil-weathered layers during the MICP process. The results indicated a positive correlation between biomass and compressive strength up to a certain point, with medium-sized particles (0.25–0.425 mm) yielding the highest compressive strength.

5.3. Further optimization of bio-bricks

The particle size distribution of the simulated soil does not follow a normal distribution. The irregular distribution of soil particles results in the uneven preparation of hollow cylinder assemblies (HCA). This variability in simulated soil distribution leads to inconsistencies in test samples, affecting their repeatability and compressive strength. To address this issue, Jiang et al. [134] developed a novel method for preparing HCAs using TJ-1 lunar soil simulant. They employed a dry bar method to prepare multilayered specimens, compacting each layer individually. The layers were then assessed to determine the highest percentage of density uniformity. The compacted layer heights for a 20 cm tall specimen with five layers were 4.2 cm, 8.4 cm, 12.5 cm, and 20 cm, respectively.

The density distribution of the sample was most uniform when these

layer heights were used. In another study, Cheng et al. [135] found that bio-bricks produced using MICP technology at 50 % bacterial saturation exhibited twice the compressive strength of those prepared under conventional fully saturated conditions, reaching a compressive strength of 9.5 MPa. Fig. 13 presents scanning electron microscope (SEM) images under 50 % saturation and 100 % saturation conditions. Calcium carbonate crystals formed at 100 % saturation exhibit a rhombohedral morphology and a size of 50 μm or larger. These crystals are in a dispersed state, with a spatial distribution characterized by scattering at the contact points and on the surfaces of sand grains.

The 'low-energy cost' advantage of MICP specifically refers to the ambient-temperature biological consolidation process and does not include the substantial upstream energy costs of reagent production, particularly calcium sourcing. Extraction of CaO from lunar regolith via pyrolysis indeed requires temperatures exceeding 1200 $^{\circ}\text{C}$ [158]. However, a systems-level analysis reveals that this energy investment is fundamentally different from that required for sintering: while sintering applies high energy directly to each individual brick, CaO production represents a bulk manufacturing process whose energy cost can be amortized over many cubic meters of construction material [67]. Furthermore, emerging technologies such as electrolytic extraction and acid leaching [159] offer potential pathways for reducing the energy footprint of calcium acquisition. Future work must focus on integrated life-cycle energy analysis and development of efficient calcium extraction methods specifically optimized for bio-cementation applications.

6. Summary and outlook

This paper reviews the current research of MICP technology for the construction of lunar and Martian bases. The following eight points summarize the challenges and the future development directions.

- (1) The Moon is rich in ilmenite, primarily in the form of TiO_2 . This material exhibits excellent photocatalytic properties, enabling electron transition on its surface under ultraviolet irradiation, which generates electron-hole pairs and promotes redox reactions. Thus, ilmenite holds the potential for hydrogen, oxygen, and water production in ISRU on extraterrestrial bodies.
- (2) Lunar or Martian soil can be strengthened by adding unique materials (such as guar gum) to improve the bonding between soil particles and enhance compressive strength. Adding glass fibers also aids in mechanical processing without requiring specific molds. However, studies indicate that excessive glass fiber content may reduce compressive strength. Future research is needed to optimize the amount of glass fiber added to minimize its negative impact on compressive strength while retaining its

benefits in reducing the dependence on specific molds for bio-bricks production.

- (3) There is ongoing research to identify superior bacterial strains that can adapt to the harsh conditions of outer space. Specifically, these strains' survival and urease production capabilities under microgravity, high vacuum, intense radiation, and extreme temperatures need to be studied. Changes in these strains at the molecular level should also be observed, or new strains may be screened for their suitability in space environments.
- (4) In bio-bricks production, applying a certain amount of pressure to pre-formed bricks can significantly enhance their compressive strength. However, whether the applied pressure should be intermittent or continuous remains unclear, as this may influence the MICP process. Furthermore, the impact of soil molding direction and the methods for controlling the stability of the applied pressure have not yet been thoroughly investigated. Developing mathematical models could provide deeper insights into these variables.
- (5) It remains a crucial factor for *Bacillus* urease activity on the Moon or Mars, where oxygen is scarce due to the thin atmosphere. Some research focus on how to sustainably produce oxygen in these environments and on understanding the impact of oxygen concentration on *Bacillus* growth. Human and livestock urine can be an auxiliary N source, supplemented with urea/urease recycling, ammonia recovery (struvite precipitation), and external or in-situ N augmentation. Efficiency can also be increased by reducing binder demand (e.g., 5–8 wt% CaCO_3), multi-round injection, and additives (fibers, polypeptides).
- (6) MICP technology can support establishing a lunar base on the Moon's surface. However, further research is required to understand how *Pasteurella* adapts and proliferates in the high-vacuum, low-gravity lunar environment. By combining 3D printing with MICP, bacteria can be embedded in a fixed carrier, which serves as a mold for shaping the material. The MICP process is activated by immersing the sample in a urea and calcium chloride cementing solution. However, the choice of carrier material may influence the urease activity and compressive strength of the bacteria. Further investigation is needed to determine how to select and process carrier materials to optimize compatibility with the bacteria.
- (7) Lunar regolith has a finer particle size, which presents a challenge when using MICP to solidify lunar soil. This fine texture may lead to low surface permeability, hindering bacterial access to the interior for mineralization. To address this, urease produced by bacteria can be released through enzyme separation, thereby enhancing urease activity and reducing the obstruction caused by

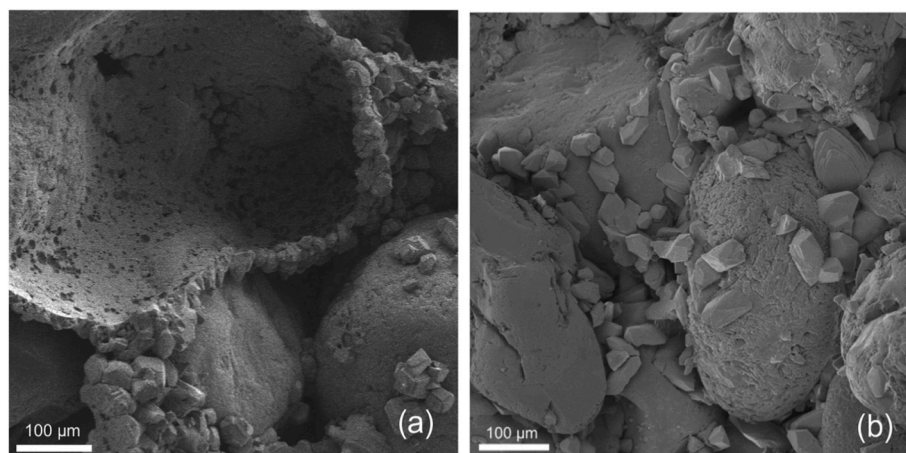


Fig. 13. SEM images of microstructure of MICP produced bio-bricks at: (a) 50 % saturation; and (b) 100 % saturation [135].

tiny pores in the lunar soil. However, the degree of bacterial fragmentation may also influence the distribution of calcium carbonate nucleation sites, requiring further optimization.

- (8) Most current experiments on manufacturing lunar building materials are conducted on Earth. Research into lunar building materials under actual lunar conditions remains underdeveloped and represents a potential direction for future study. Scholars plan to send space bricks aboard the "Tianzhou 8" cargo spacecraft to the Chinese space station for testing. The aim is to explore whether these space bricks can be used to construct habitats on the Moon. Upon completion of the space experiments, the space bricks are expected to return to Earth by the end of 2025.

CRediT authorship contribution statement

Wenjing Yang: Investigation, Writing – original draft, Writing – review & editing, Visualization. **Yuqi Song:** Conceptualization, Writing – review & editing, Supervision, Funding acquisition. **P.G. Ranjith:** Supervision, Writing – review & editing. **Zhenqian Chen:** Resources. **Hao Fang:** Writing – review & editing. **Derek Elsworth:** Writing – review & editing. **Fengshou Zhang:** Writing – review & editing. **Youzhi Feng:** Writing – review & editing. **Xiangyang Song:** Conceptualization, Methodology, Resources, Writing – review & editing, Supervision, Funding acquisition, Project administration.

Consent to participate

The authors declare that they consent to participate.

Consent for publication

The authors declare that they consent to publication.

Availability of data and material

Not applicable.

Code availability

Not applicable.

Ethics approval

This article does not contain any studies with human participants or animals performed by any of the authors.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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